

Flybot: A Dual Active Wheel Hybrid Land-Air Robot With Five-Link Leg Joints

Hongyu Xu^{ID}, Xiangming Zheng^{ID}, Yu Wang^{ID}, and Luwei Liao^{ID}

Abstract—Small land-air robots with rapid response and field operation capabilities hold significant potential for emergency rescue and field exploration applications. However, achieving stable and efficient mode transitions, along with effective terrain adaptability, remains a challenge. This letter presents a novel land-air robot, Flybot, which combines a bicopter UAV with a dual active-wheel, self-balancing wheel-legged base. Unlike traditional air-ground robots, this paper designs a single-drive, five-link leg structure that enhances lateral stability and terrain adaptability. By optimizing the five-link joint, we minimize energy consumption in ground mode, resulting in improved efficiency. Additionally, this paper establishes a hybrid dynamics model to describe Flybot's mode transitions and introduces a novel hybrid transition controller that leverages acceleration and throttle data to determine transition states. Finally, the adaptability and reliability of the robot are verified by the ground obstacle crossing experiment and the air-ground continuous motion experiment in the field, and the effectiveness of the hybrid transition controller of the robot is demonstrated. The weight-to-power ratio of the Flybot in air mode is 5.18 g/W, and the weight-to-power ratio in ground mode is 309 g/W, indicating the high efficiency of the Flybot.

Index Terms—Aerial systems; mechanics and control, filed robots, mechanism design, legged robots.

I. INTRODUCTION

IN RECENT years, multi-rotor UAVs have been widely used in aerial photography, mapping, reconnaissance, search and rescue, etc. due to their high flexibility and strong load capacity [1], [2], [3], especially in some of the more dangerous areas that are difficult for humans to reach [4], [5], [6]. However, it also faces problems such as single function, high energy consumption and limited operating space. In certain situations where ground interaction is necessary, a single UAV cannot accomplish such tasks, therefore, to expand the versatility of the

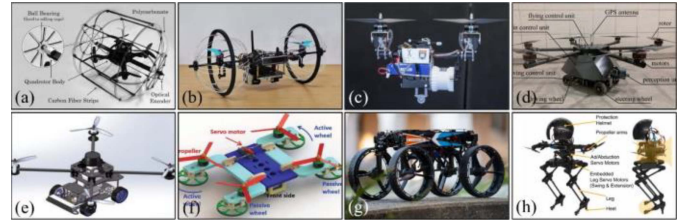


Fig. 1. Existing typical land-air amphibious equipment.

UAV, one of the effective ways is to add additional motion modes to them [7]. For example, a land motion module can be combined with a multi-rotor, thus taking advantage of both strengths. At present, the development of hybrid land-air multimodal robots can be divided into four main categories: passive wheeled, active wheeled, variable structure, and legged [8].

In the research of passive multimodal robots, there is the addition of a free-rolling rolling cage to the periphery of a multirotor UAV [8] (as shown in Fig. 1(a)) or adding passive ball wheels on both sides [9] (shown in Fig. 1(b)) and passive rollers [7], [10], also as shown in the article [11] (as shown in Fig. 1(c)), a single passive wheel is installed under the multirotor to extend the UAV's ability to move on the ground. Although this design avoids the need for additional actuators to facilitate ground motion, its passive nature results in limited energy efficiency, as it necessitates a continuous supply of kinetic energy from the rotor motors. Moreover, its functionality is constrained to movement on flat surfaces.

Active-wheeled multimodal robots typically integrate four-wheeled vehicles with multi-rotors. Ground locomotion is achieved through wheeled mechanisms, which commonly require a dedicated steering system. However, this approach often results in increased volume and a large turning radius, thereby limiting the robot's maneuverability and overall flexibility. For example, in [12] (as shown in Fig. 1(d)), the combination of Ackermann chassis and hexacopter is used to achieve continuous air-to-ground movement. Similarly, in article [13] (shown in Fig. 1(e)), the configuration of quadcopter and unmanned vehicle can also achieve continuous air-land movement. The Double-Bee designed in [14] is most similar to the design proposed in this paper, with a dual active wheel and a dual-rotor configuration, which realizes the decoupling of pitch direction through vector thrust. However, such designs are typically restricted to flat roads, with poor stability in the transverse roll direction, and have limited obstacle-crossing capabilities. While the addition of new power modules can improve ground stability and enhance

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energy efficiency, it also increases the size of the vehicle, thereby constraining its mobility and flexibility.

Some researchers have introduced novel mechanisms to multi-rotor systems to enable terrestrial movement through structural modifications. For instance, adding a stretching mechanism to realize four rotors with adjustable stretching angles [15] (shown in Fig. 1(f)), which can be changed to a four-wheeled mode on the ground to move through the outer rims. Similarly, there are also machines like the Morphobot [16] (shown in Fig. 1(g)) that can achieve a variety of deformations, which can cope with more terrains. In [17], [18], a robot capable of ground locomotion through structural deformation is proposed. While this design enables terrestrial mobility, the excessive use of hinge mechanisms significantly restricts the robots' load capacity. Additionally, the structural deformation limits the robot's maneuverability.

In addition, some researchers have combined gait robots with multi-rotors to design a novel land-air robot [19] (Fig. 1(h)), which not only walks on the ground but also performs tightrope walking, skateboarding, etc., and exhibits strong stability performance. Similarly, a dual-rotor flight module [20] has been added to the humanoid robot to make it capable of flying. It is well known that anthropomorphic gait walking is inefficient and less flexible, so the application prospect still has some limitations. In [21], researchers developed a novel land-air robot that integrates flapping wings, rotors, and adhesive materials. This design enables the robot to climb and remain stationary on vertical surfaces, showing a unique capability for versatile locomotion.

The contributions of this work are as follows:

- 1) A land-air robot featuring a single-drive five-link joint was designed, with its power consumption optimized to enhanced the efficiency of the robot.
- 2) A hybrid transition controller was developed based on the dynamic balance characteristic of pitching, enabling the robot to achieve stable mode transitions in complex environments.
- 3) A critical state mode transition control strategy, utilizing acceleration and throttle inputs, was proposed to facilitate a real-time and efficient transition process.
- 4) Field experiments and indoor physical demonstrations were conducted to validate the system's feasibility.

The rest of this letter is organized as follows. Section II describes the general layout, structure design, and critical components of Flybot in detail. Section III presents the dynamic model of the robot's hybrid transition mode and the optimization of the five-link joint design. Section IV focuses on the design of the control strategy, while Section V discusses the experimental tests and the analysis of the results. Finally, Section VI concludes the letter with a summary and outlines directions for future work.

II. SYSTEM DESIGN

Flybot consists of a fuselage and two functional parts: a flight unit and a ground motion unit. The flight unit is a conventional tilt-rotor bicopter UAV configuration, while the ground motion unit is comprised of two single-drive, five-link wheel legs. These

TABLE I
FLYBOT PART PARAMETERS

Component		Num	Type	Weight(g)
Flight Part	Rotor Motor	2	T-Motor F90 1300KV	46.6
	Tilt Servo	2	GDW RS0708	20
	ESC	1	Hobbywing 65A 4-IN-1	15
Control System	Flight Controller	1	Pixhawk6C MINI	39.2
Dynamical System	Battery	1	Tattu 6S 1050mAh	170
Land Part	Wheel Leg Motor	2	DJI M2006 P36	90
	Joint Servo	2	KST A20-4515	80
	ESC	2	DJI C610	17

legs are designed to streamline the system's structure, minimize the number of actuators, and reduce the robot's size and weight. The dual active wheels are driven by brushless motors, which communicate with the flight control via a CAN bus.

The design requirement for the wheel and leg structure is that the height of obstacle crossing is not less than 50 mm, which, on the one hand, can cross the obstacles by lifting and lowering the joints when moving on the ground. On the other hand, it can ensure the stability of the rolling attitude when traveling on an unstructured road surface to enhance the passability of the robot. The joint servo drives the shoulder joint gear set to generate the reciprocating motion of the rocker, and the wheel-leg motors are connected to the two calf connecting rods, which in turn are connected to the two rockers, thus realizing the lifting and lowering motion of the wheel-leg joints. Finally, the maximum height of the robot is 294 mm, the maximum width is about 300mm, the distance between the front and back of the body is about 140 mm, and the weight of the whole robot is 1.44 kg. Table I gives information on all the components of Flybot and their parameters. The detailed structure is shown in Fig. 2.

III. MODELING

In this letter, the coordinate system is defined as follows: $\mathcal{F}^W : \{o^W, x^W, y^W, z^W\}$ denotes the NED (north-east-down) inertial coordinate system and is denoted by W in the upper right corner. $\mathcal{F}^B : \{o^B, x^B, y^B, z^B\}$ denotes the body coordinate system, the upper right corner label is denoted by B to distinguish it from the inertial coordinate system, the wheel coordinate system is denoted by $\mathcal{F}_i^M : \{o_i^M, x_i^M, y_i^M, z_i^M\}$, the rotors coordinate system is denoted by $\mathcal{F}_i^R : \{o_i^R, x_i^R, y_i^R, z_i^R\}$, o is the origin of the coordinate system, and xyz follows the right-hand rule. $R_a^b(\bullet)$ denotes the rotation matrix from frame a to frame b , φ is roll Euler angle, θ is pitch Euler angle, and ψ is yaw Euler angle. The detailed coordinate system is given in Fig. 3.

A. Aerial Mode and Ground Mode

In aerial mode, the Flybot operates as a bicopter UAV, similar to the traditional bicopter UAV model, which has been described in detail in [23]. In ground mode, it transitions into a balanced robot, with a more comprehensive description of the modeling

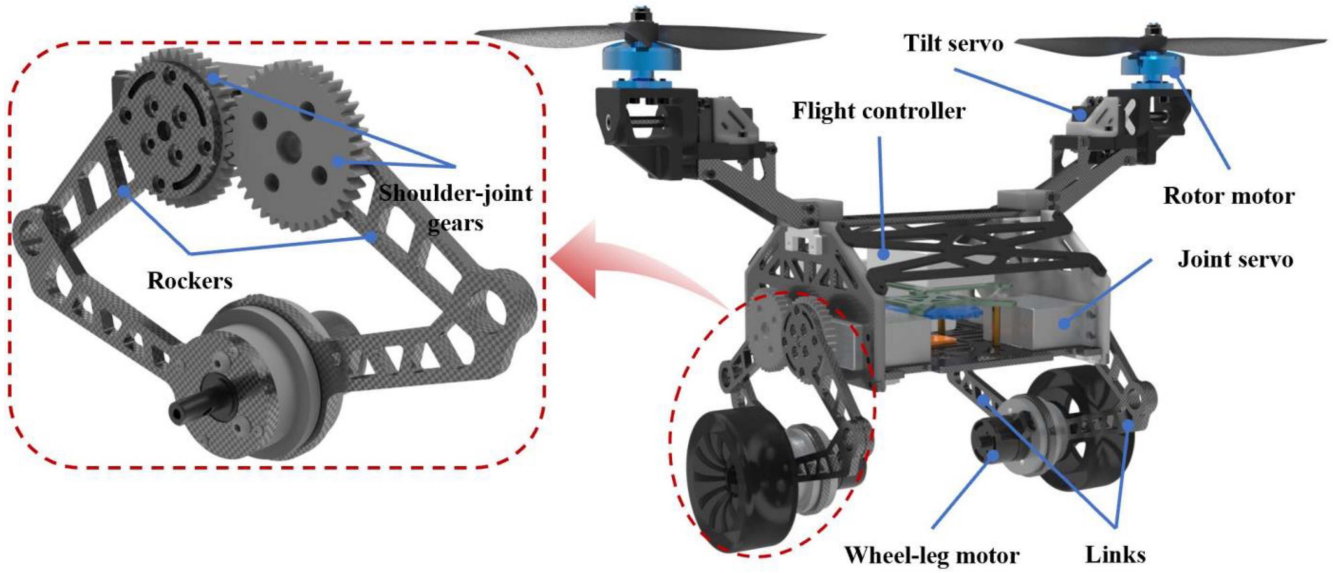


Fig. 2. Flybot 3D model diagram and parallel leg structure diagram

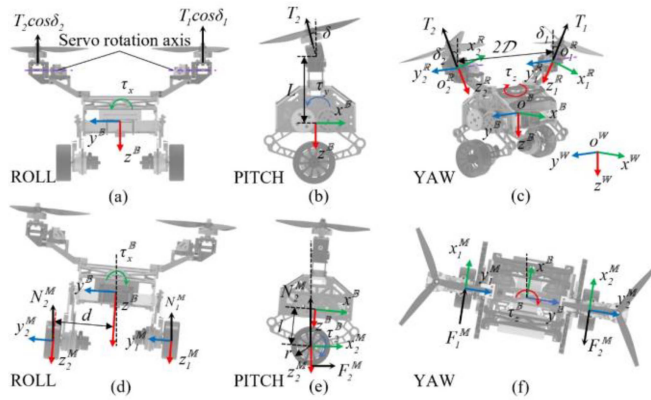


Fig. 3. The coordinate system of the Flybot. (a) Aerial mode roll control. (b) Aerial mode pitch control. (c) Aerial mode yaw control. (d) Ground mode roll control. (e) Ground mode pitch control. (f) Ground mode yaw control.

provided in [14]. As such, we will not elaborate further on these models here.

B. Hybrid Transition Mode

When the robot is in hybrid transition mode, it is characterized by the tiltrotor acting in conjunction with the balanced motion part, and the dynamics model of the robot is given by (1):

$$\begin{cases} M\ddot{\mathbf{P}}^W = \mathbf{f}_G^W + \mathbf{R}_B^W \sum_{i=1}^2 \mathbf{T}_i^B + \mathbf{R}_M^W \sum_{i=1}^2 (\mathbf{N}_i^M + \mathbf{F}_i^M) \\ \mathbf{J}^B \dot{\boldsymbol{\omega}}^B = \mathbf{R}_M^B \sum_{i=1}^2 (\mathbf{d}_i^M \times \mathbf{F}_i^M) + \mathbf{R}_R^B \sum_{i=1}^2 (\mathbf{d}_i^R \times \mathbf{T}_i^R) \\ -\boldsymbol{\omega}^B \times \mathbf{J}^B \boldsymbol{\omega}^B \end{cases} \quad (1)$$

where M and $\mathbf{f}_G^W$ represent the mass and gravity vector of the robot in the frame $\mathcal{F}^W$, and $\mathbf{f}_G^W = [0 \ 0 \ mg]^T$, $\mathbf{P}^W$ represents the displacement vector of the robot's centroid. $\mathbf{T}_i^R$ represents the pull vector generated by the propeller and in the hovering

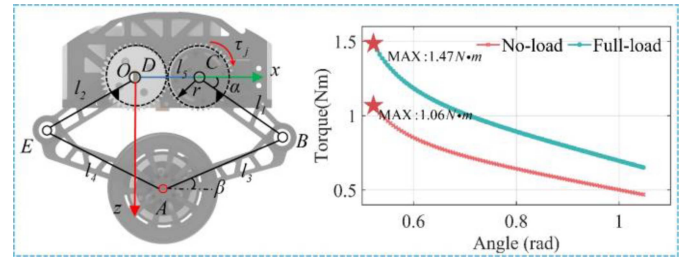


Fig. 4. Schematic diagram of five-link leg mechanism and joint torque variation diagram.

state, $\mathbf{T}_i^R = [0 \ 0 \ -T_i \cos \delta_i]^T$. $\mathbf{F}_i^M$ and $\mathbf{N}_i^M$ are the friction force acting on the wheel and the support force from the ground, respectively. $\mathbf{d}_i^R$ and $\mathbf{d}_i^M$ represent the distance vector from the body's center of mass (CoM) to the rotor's CoM, and from the point of friction force application to the wheel's CoM, respectively. $\mathbf{J}^B$ and $\boldsymbol{\omega}^B$ are the inertial matrix and the angular velocity vector of the robot, respectively.

C. Modeling of Five-Link Leg Structure

The closed-chain five-link mechanism is widely utilized in robotics due to its stability and flexibility. Based on this mechanism, a set of shoulder joint gears is added to the wheel leg joints in this letter. This design reduces the number of actuators, simplifies the system's control architecture, and contributes to the reduction of the Flybot's size and weight.

The mechanism diagram of the five-link joint is given in Fig. 4, wherein the rod l_5 represents the body, l_1 and l_2 are the active and driven rods, respectively, and l_3, l_4 are two lower leg connecting rods. r is the radius of the indexing circle of the gear, and $r = l_5/2$. α is the angle between l_1 and the positive of the X-axis ($\alpha \in [\pi/6, \pi/3]$), and β is the angle between l_3 and the horizontal to the right. The position of the point A in the center of the wheel is (x_A, z_A) , and the following formula gives the position of A

in the vertical direction:

$$z_A = l_1 \sin \alpha + l_3 \sin \beta \quad (2)$$

where $\beta = \arccos((r + l_1 \cos \alpha)/l_3)$. From this, the acceleration at point A can be calculated as follows:

$$\ddot{z}_A = l_1(\ddot{\alpha} \cos \alpha - \dot{\alpha}^2 \sin \alpha) + l_3(\ddot{\beta} \cos \beta - \dot{\beta}^2 \sin \beta) \quad (3)$$

Based on the Newton-Euler method, the dynamic equation of the closed-chain five-link mechanism is derived, and the relationship between the joint servo torque τ_j , the link lengths l_1, l_3 and angle α is expressed as follows:

$$\tau_j = \frac{G \sin(\alpha + \beta) \bullet l_1}{\sin \beta} \quad (4)$$

where $G = 1/2Mg$.

D. Optimized Design of Five-Link Joint Mechanism

From the dynamic equation, it can be observed that the active torque at the joint is influenced by the swing angle α and the lengths of the rod l_1, l_3 and l_5 . Therefore, the optimization target parameters can be defined as follows:

$$\begin{cases} \text{minimize } f(\mathbf{X}) = \tau_j \\ \text{S.t. } 0.05 \leq l_1 \leq 0.06 \\ 0.05 \leq l_3 \leq 0.07 \\ 0.02 \leq r \leq 0.03 \\ g(x_1) = l_1 \sin \alpha_{\min} + l_3 \sin \beta_{\min} - 0.045 \leq 0 \\ g(x_2) = 0.04 + l_1 \sin \alpha_{\min} + l_3 \sin \beta_{\min} - l_1 \sin \alpha_{\max} \\ + l_3 \sin \beta_{\max} \leq 0 \end{cases} \quad (5)$$

Using the `fmincon` function of the *MATLAB* optimization toolbox, we can get the final result: $r = 0.02$ m, $l_1 = 0.05$ m, and $l_3 = 0.065$ m. The resulting torque-swing angle relationship is shown in Fig. 4, where the red line represents the torque output at no load, and the blue line represents the output of torque under full load. The maximum torque required at the minimum swing angle is $1.47 \text{ N} \cdot \text{m}$, which serves as a key reference for selecting the joint drive.

IV. CONTROL DESIGN

This section will develop the control model of Flybot for its three operational modes: aerial mode, ground mode, and transition mode.

A. Aerial Mode

In aerial mode, Flybot operates as a bicopter UAV. Its control strategy follows the standard approach used in most traditional bicopter UAVs, employing a cascade PID controller to regulate both the aircraft's position and attitude.

1) *Position and Attitude Controller*: For the position controller, the desired input is $\mathbf{Q}_d = [(\mathbf{P}_d^W)^T \ \psi_d]^T$, the velocity error can be obtained as follows:

$$\mathbf{e}_v = \dot{\mathbf{P}}_d^W - \dot{\mathbf{P}}^W \quad (6)$$

The desired acceleration calculated by the cascade PID controller is as follows:

$$\mathbf{a}_d^W = K_P^v \mathbf{e}_v + K_D^v \dot{\mathbf{e}}_v + K_I^v \int \mathbf{e}_v \quad (7)$$

where K_P^v , K_D^v and K_I^v are proportional, differential, and integral gains, respectively. According to the method in [22], the desired attitude matrix and desired force can be obtained from the following equation:

$$M \mathbf{a}_d^W = \mathbf{f}_G^W + \mathbf{R}_d \mathbf{T}_d^B \quad (8)$$

The attitude controller tracks the attitude error vector $\mathbf{e}_r$ to obtain the desired angular velocity and then tracks the angular velocity error $\mathbf{e}_\omega$ to obtain the desired moment $\mathbf{M}_d^B$:

$$\begin{aligned} \omega_d^B &= K_P^r \mathbf{e}_r \\ \mathbf{e}_\omega &= \omega_d^B - \omega^B \end{aligned} \quad (9)$$

$$\mathbf{M}_d^B = K_P^\omega \mathbf{e}_\omega + K_D^\omega \dot{\mathbf{e}}_\omega + K_I^\omega \int \mathbf{e}_\omega \quad (10)$$

where K_P^r is the proportional term of the outer loop, K_P^ω , K_D^ω and K_I^ω are the proportional, differential, and integral term of the inner loop, and the desired moment $\mathbf{M}_d^B = [\tau_{xd}^B \ \tau_{yd}^B \ \tau_{zd}^B]^T$.

2) *Control Allocator*: Referring to the method in [23], rewrite the input of the system as $\mathbf{u} = [u_1 \ u_2 \ u_3 \ u_4]^T$, which can be represented by the thrust vector as follows:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} T_1 \sin \delta_1 \\ T_1 \cos \delta_1 \\ T_2 \sin \delta_2 \\ T_2 \cos \delta_2 \end{bmatrix} \quad (11)$$

where $u_1 = \tau_{yd}^B/L$, $u_2 = \tau_{zd}^B/D$, $u_3 = \tau_{xd}^B/D$ and $u_4 = -T_{zd}^B$.

Therefore, the thrust and moment vector of the system can be expressed by:

$$\begin{aligned} \mathbf{T}^B &= [0 \ 0 \ -u_4]^T \\ \mathbf{M}^B &= [D \cdot u_3 \ L \cdot u_1 \ D \cdot u_2]^T \end{aligned} \quad (12)$$

According to the expected z -axis thrust T_{zd}^B and the desired moment $\mathbf{M}_d^B$, the rotor thrust T_i and servo angle δ_i can be obtained from (12):

$$T_1 = \frac{1}{2} \sqrt{(u_1 - u_2)^2 + (u_3 + u_4)^2} \quad (13)$$

$$T_2 = \frac{1}{2} \sqrt{(u_1 + u_2)^2 + (u_4 - u_3)^2} \quad (14)$$

$$\delta_1 = \text{atan} \left(\frac{u_1 - u_2}{u_3 + u_4} \right) \quad (15)$$

$$\delta_2 = \text{atan} \left(\frac{u_1 + u_2}{u_4 - u_3} \right) \quad (16)$$

B. Ground Mode

In ground mode, Flybot operates as a retractable self-balance robot. The body balance is maintained through a cascade PID control system, comprising an upright loop and a velocity loop.

The velocity loop tracks the wheel speed error and outputs the desired pitch angle θ_{gd} as its output:

$$\theta_{gd} = K_P^v e_v + K_I^v \int e_v \quad (17)$$

where K_P^v and K_I^v are the proportional and integral terms of the velocity loop, respectively. The upright loop tracks the pitch angle error and outputs the desired pitch angular velocity $\omega_{\theta d}^B$, and then tracks the pitch angular rate error to obtain the desired control torque τ_t :

$$\omega_{\theta d}^B = K_P^\theta e_\theta + K_D^\theta \dot{e}_\theta \quad (18)$$

$$\tau_t = K_P^\omega e_\omega + K_D^\omega \dot{e}_\omega \quad (19)$$

where K_P^θ , K_D^θ and K_P^ω , K_D^ω are the proportional and differential terms of the angular loop and the pitch angular velocity loop, and e_θ , e_ω denote the pitch angle error and pitch angular velocity error. Additionally, the turn loop tracks the yaw rate error e_ω^ψ :

$$\begin{aligned} \tau_l &= \tau_t - (K_P^\psi e_\omega^\psi + K_D^\psi \dot{e}_\omega^\psi) \\ \tau_r &= \tau_t + (K_P^\psi e_\omega^\psi + K_D^\psi \dot{e}_\omega^\psi) \end{aligned} \quad (20)$$

where τ_l and τ_r are the control torque of the final output of the left and right wheel-foot motors. The error e_ω^ψ is the yaw angular velocity error, while K_P^ψ and K_D^ψ are the proportional and differential terms of the turn loop.

The rolling attitude of the Flybot during ground motion is controlled by the joint servos, utilizing a cascade controller. The inner loop tracks the roll angle rate error e_φ^ω , while the outer loop tracks the roll angle error e_φ :

$$\omega_{\varphi d}^B = k_p^\varphi e_\varphi \quad (21)$$

$$\tau_j = K_P^\varphi e_\varphi + K_D^\varphi \dot{e}_\varphi \quad (22)$$

where τ_j is the control torque of the joint servo, k_p^φ is the proportional gain of the outer loop, and K_P^φ and K_D^φ are the inner loop PD controller's proportional and differential terms, respectively.

C. Transition Mode

Flybot exhibits good robustness as a bicopter UAV. However, during landing, the dynamic balance characteristics in the pitching direction make it susceptible to disturbances, which can result in oscillations, diverge or even tipping over. Therefore, it is essential to design an appropriate controller to ensure landing stability.

During landing, the rotor thrust, in conjunction with the active wheels, functions as the system's recovery force to enhance stability in the pitch direction. Consequently, the new desired pitch angular velocity $\omega_{\theta d}^B$ can be obtained using the following formula:

$$\omega_{\theta d}^B = \lambda_1 \omega_{\theta d}^B + \lambda_2 \omega_{\theta d}^B \quad (23)$$

where $\omega_{\theta d}^B$ represents the desired pitch angular velocity of the bicopter UAV, λ_1 and λ_2 are the two impact factors ($\lambda_1, \lambda_2 \in$

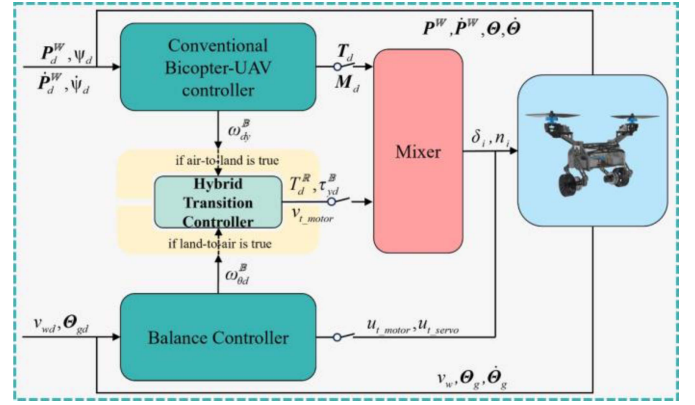


Fig. 5. The control structure of Flybot.

$(0, 1)$). It can be found that, when $\lambda_1 = 1, \lambda_2 = 0$, this is the pitch attitude controller for the bicopter UAV, and when $\lambda_1 = 0, \lambda_2 = 1$, this is the pitch attitude controller for the balanced robot. The new pitch angular velocity error $\tilde{e}_{wy}$ and $\tilde{e}_\omega^\theta$ can be obtained by $\omega_{\theta d}^B$, therefore, (10) and (19) can be rewritten as:

$$\tau_{yd}^B = k_P^\omega \tilde{e}_{wy} + k_D^\omega \dot{\tilde{e}}_{wy} + k_I^\omega \int \tilde{e}_{wy} \quad (24)$$

$$u_w = K_P^\omega \tilde{e}_\omega^\theta + K_D^\omega \dot{\tilde{e}}_\omega^\theta \quad (25)$$

where τ_{yd}^B indicates the desired pitching moment. Finally, the updated outputs for the rotors, servos, and motors are computed. The hybrid transition controller integrates the attitude controller of the dual-rotor UAV with the attitude controller of the balance robot, and the same components are omitted here. By tuning the two weight factors, different transition effects can be achieved. The control structure of the Flybot is illustrated in Fig. 5.

D. Transition Strategy

1) *Land-to-Air Transition*: When Flybot is prepared to transition into aerial mode, the process begins with the pilot issuing a take-off command (unlock). Following this, the system receives throttle σ input from the pilot and judging the entry into the aerial mode by detecting the robot's acceleration in the Z-direction at the moment it detaches from the ground. We define a conversion factor κ to describe the land-to-air transition process. This process can be mathematically described using the following piecewise function:

$$\kappa = \begin{cases} 0, & \sigma \geq 0.5, |a_z| \geq a_{c1} \\ 1, & \sigma \leq 0.3, |a_z| > a_{c2} \end{cases} \quad (26)$$

where a_z is the acceleration along the Z-axis, a_{c1} and a_{c2} are the acceleration constant in the Z-direction during takeoff and landing, respectively. The smoothness of the transition process can be regulated by adjusting their values.

The output of all actuators of Flybot is represented by U as follows:

$$U = \kappa \cdot U_1 + (1 - \kappa) \cdot U_2 \quad (27)$$

where U_1 and U_2 are the outputs of the actuators in ground mode and air mode. After entering the air mode, the actuators in the

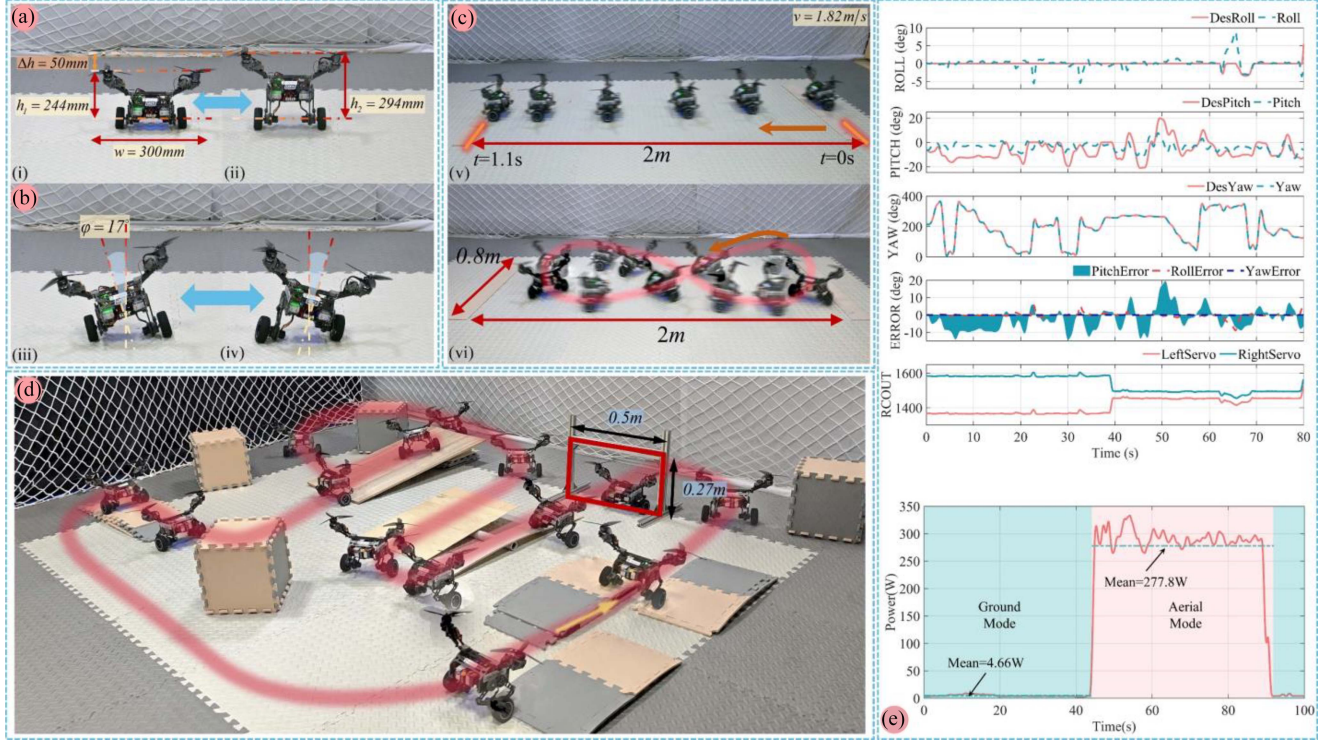


Fig. 6. Indoor obstacle crossing experiments. (a) Joint height regulation movement. (b) The robot tilts sideways from side to side. (c) Straight-line fast movement on flat road surface and figure of eight fast movement. (d) Motion stability experiments in complex environments. (e) Flybot's roll, pitch, yaw, angular error, joint servo output in ground mode, and power consumption in land-air cross-modal motion.

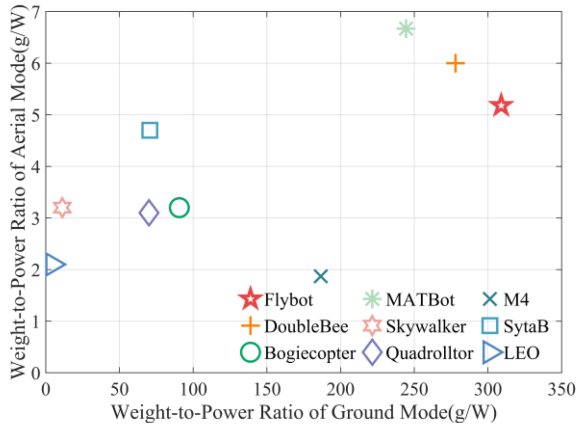


Fig. 7. The power efficiency of Flybot and other typical land-to-air robots.

ground mode cease their output, in this case, $U = U_2$, indicating that the robot is fully operating as a bicopter UAV.

2) *Air-to-Land Transition*: First, the S -shaped function is introduced:

$$S_w = 1 - \frac{1}{1 + e^{-(x-B)/R}} \quad (28)$$

where B is the parameter of the control function offset. In this letter, B represents the amount of throttle input by the pilot, and R is the parameter of the control function's slope. We can change the values of these two parameters to achieve the desired effect.

When it needs to enter the land mode, the pilot first issues the landing command S_t . Once $S_t = 1$, the robot enters the landing

detection mode. At this time, as the throttle input from the pilot gradually decreases, the influence factor S_w of the joint servos gradually increases. Once the touchdown conditions are detected as satisfied, the robot is controlled by the hybrid pitch controller, and after the attitude is stabilized, $U = U_1$, indicating that the robot has fully transitioned into a balance robot. At this point, the actuators used in aerial mode cease their output, and the system operates entirely in ground mode.

V. EXPERIMENT VALIDATION

In this section, we will validate the reliability and adaptability of the Flybot in complex environments through a series of experiments. Additionally, we will assess the feasibility of the mode transition control strategy and the pitch controller by conducting land-to-air mode transition experiments.

A. Environmental Adaptability and Mobility

We first demonstrated the joint adjustment performance of the Flybot. The maximum lifting of the wheel leg joint is 50 mm, as shown in Fig. 6(a), providing a significant advantage in navigating narrow spaces. By adjusting the height of the left and right joints, a maximum roll angle of 17° can be achieved without tipping, enabling stable ground movement, as depicted in Fig. 6(b). Next, the maneuverability of the Flybot was tested through a straight-line rapid maneuver and a figure-of-eight maneuver, shown in Fig. 6(c). The results indicated that the maximum speed on flat ground was 1.82 m/s, which surpasses the maximum speed of 1.35 m/s observed in the MTABot and

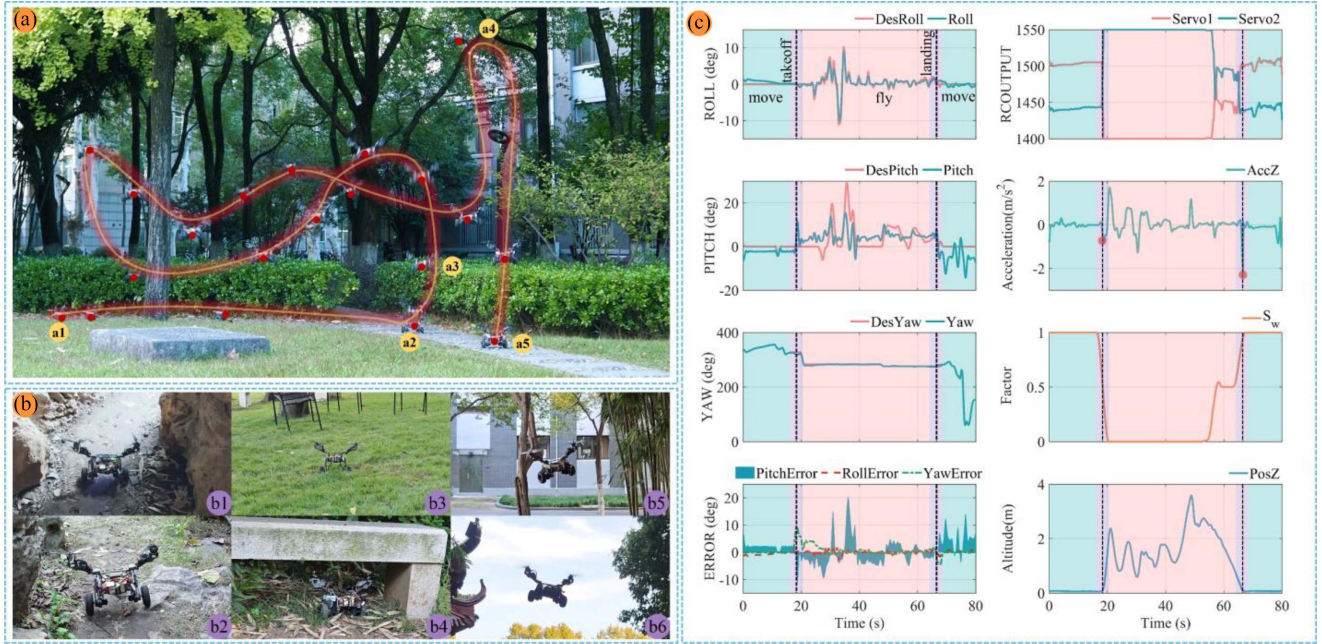


Fig. 8. Move in complex outdoor environments. (a) Modal transition experiments. (b) In the field of complex environment experiments, Flybot can move in caves, grass, and other scenes by adjusting the height of the joints to pass through low and narrow areas; due to its small size, it can also fly in the jungle. (c) A complete cross-modal motion process data diagram.

the speed of the DoubleBee was unknown and could not be compared. Finally, indoor testing revealed that the Flybot could easily traverse various obstacles, such as single-plank bridges and slopes, as shown in Fig. 6(d). These results indicate that the Flybot with a five-link joint has a clear advantage in ground passability.

The pitch angle, roll angle, yaw angle, angle error, and joint servo output data of the robot during ground motion are shown in Fig. 6(e). It can be observed that due to the five-link joint, the robot's rolling attitude is relatively stable when moving on the ground, the roll angle error is small, and the yaw angle of the ground movement is also stable. However, due to the characteristics of dynamic balance, the overall pitch angle error is large, which highlights an area for future optimization.

B. Power Consumption

Through experiments, it was found that the average power consumption of the Flybot is only 4.66 W when operating in low-power consumption ground mode, while in aerial mode, the average power consumption increases 277.8 W, as shown in Fig. 6(e). The weight-to-power ratio is approximately 309 g/W in ground mode and 5.18 g/W in aerial mode. When compared to several typical land-air robots, such as MTABot [24], M4 [16], DoubleBee [14], Skywalker [25], SytaB [9], BogieCopter [26], Quadrolltor [27], and LEO [19], Flybot's weight-to-power ratio in aerial mode is slightly lower than that of the MTABot and DoubleBee. However, its weight-to-power ratio in ground mode is the highest, as shown in Fig. 7. These results demonstrate the effectiveness of the Flybot and show that the robot's ability to switch between modes based on the working environment can significantly extend its operational time, thereby better meeting

diverse task requirements. This capability is one of the promising directions for future research.

C. Field Exploration and Modal Switching

We tested the Flybot outdoors in a more complex environment, and its smaller size, combined with the adjustable five-link leg joints, enabled it to navigate through the jungle terrain, as shown in Fig. 8.

In Fig. 8(a), the Flybot moves from point a1 to point a2 along a stony pavement. Upon reaching a2, it executes a land-air transition command and transitions into the aerial mode, flies through the woods, and finally lands back on the pavement at point a5, completing the air-to-land transition. In Fig. 8(b), b1 and b2 illustrate the Flybot navigating through a narrow cave, and b3 shows the Flybot moving across the grass while adjusting the height of its body via joint mechanisms to enhance its mobility. In b4, the Flybot lowers its body height to pass under a low stone chair, and in b5 and b6, the robot flies through the bamboo forest. These experiments, conducted in complex environments, demonstrate that the Flybot exhibits robust adaptability and stability, highlighting its potential for diverse operational scenarios.

The data from Flybot's outdoor experiments are shown in Fig. 8(c), and it can be found that the roll angle and yaw angle of the robot are always relatively stable, and the error is small. During take-off, a significant change in Z-axis acceleration is observed, indicating that the Flybot has transitioned into the aerial mode. Concurrently, the output curves of the joint servo actuators, Servo1 and Servo2, become a straight line, signifying that the joint servos cease functioning in aerial mode. Upon landing, the Z-axis acceleration undergoes another noticeable

change as the robot makes ground contact. Prior to touchdown, as the pilot reduces throttle input, the influence factor S_w of the joint servo follows an S-shaped curve, demonstrating a gradual increase in servo output to adapt to the terrain. This continuous land-air transition process validates the effectiveness of the Flybot's transition control strategy and pitch controller, ensuring seamless transition across different modes.

VI. CONCLUSION

In the work of this letter, we propose a land-air robot Flybot with a combination of five-link joints, a dynamically balanced two-wheel system, and a bicopter UAV, which can not only move smoothly and quickly in complex terrain but also fly stably in the air. In addition, we propose a new multi-drive fusion mode transition controller based on acceleration and throttle data to judge the state of the robot, which can judge the state of the robot in real-time and efficiently and achieve a smooth transition between the two modes. The experimental results show that Flybot has strong adaptability in complex environments, can easily cope with caves, grasslands, stone roads, etc., in field exploration, and can smoothly realize the mode transition in complex environments in the field under the action of a hybrid transition controller. Compared with most active wheeled land-to-air robots, the configuration of Flybot's twin rotors and dual driving wheels makes it smaller, which brings the advantages of fast-moving speed, small steering radius, and the ability to maneuver in narrow environments. In conclusion, the Flybot and hybrid transition controller we designed have certain reference significance for the future study of small land-air robots moving in complex environments.

In future work, we will explore the impact of additional environmental disturbance factors on Flybot and investigate more advanced control methods, such as *RL* and *MPC*, to further enhance system stability and reliability. One of our ultimate objectives will be achieving autonomous navigation and mode switching for Flybot in unknown environments.

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